NETWORK TECHNOLOGY
UNIT- 3

- ❖ **Wireless Ad Hoc Networks :** "Ad hoc" means makeshift or improvised, so a wireless ad hoc network (WANET) is a type of on-demand, impromptu device-to-device network. In ad hoc mode, you can set up a wireless connection directly to another computer or device without having to connect to a any infrastructure.
  - ➢ **Types of Wireless Ad Hoc Networks**
    - Mobile ad hoc network (MANET)
    - Vehicular ad hoc network (VANET)
    - Smartphone ad hoc network (SPANC)
    - Flying Adhoc network(FANET)

❖ **Introduction of Mobile Ad hoc Network (MANET)**

- MANET stands for Mobile Adhoc Network also called a wireless Ad Hoc network or Ad Hoc wireless network ..
- A MANET consists of a number of mobile devices that come together to form a network as needed, without any support from any existing internet infrastructure or any other kind of fixed stations.
- They consist of a set of mobile nodes connected wirelessly in a **self-configured, self-healing network** without having a fixed infrastructure.
- MANET nodes are free to move randomly as the network topology changes frequently. Each node behaves as a router as they forward traffic to other specified nodes in the network.

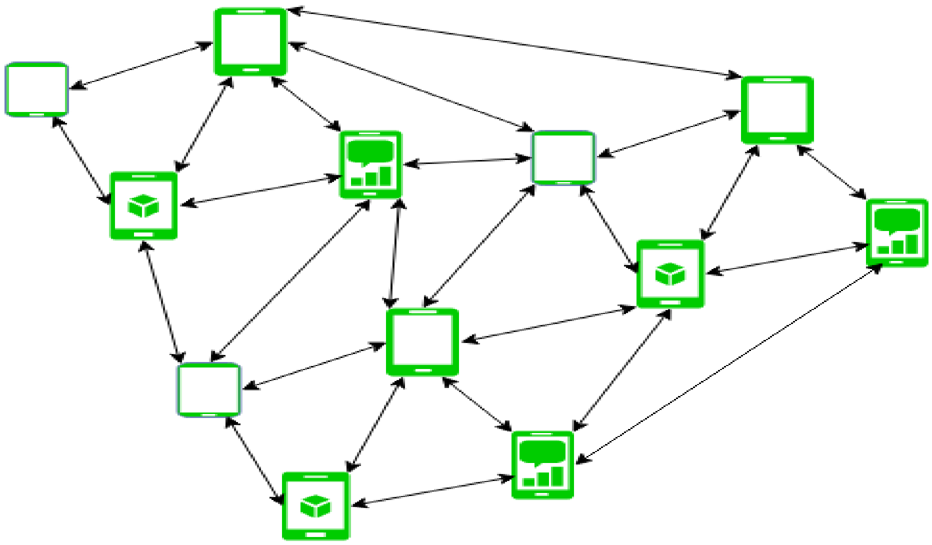


Figure - Mobile Ad Hoc Network

❖ Characteristics of MANET

- **Dynamic topologies:** nodes are free to move arbitrarily; thus the network topology may be changed randomly and unpredictably and primarily consists of bidirectional links. In some cases where the transmission power of two nodes is different, a unidirectional link may exist.
- **Bandwidth constrained, variable capacity links:**
  Wireless links usually have lower reliability, efficiency, stability, and capacity as compared to a wired network
- **Energy Constrained Operation:**
  As some or all the nodes rely on batteries or other exhaustible means for their energy. Mobile nodes are characterized by less memory, power, and lightweight features.

- **Limited physical security:** MANETs are generally more having a tendency to physical security threats than wireline networks.The increased possibility of denial of services (DoS) attacks should be considered carefully. To reduce security threats, many existing link security techniques are often applied within wireless networks.
- **Less Human Intervention:**
  They require minimal human intervention to configure the network, therefore they are dynamically autonomous in nature.

Advantages:

Flexibility: MANETs are highly flexible, as they can be easily deployed in various environments and can be adapted to different applications and scenarios. This makes them ideal for use in emergency situations or military operations, where there may not be a pre-existing network infrastructure.

Scalability: MANETs can easily scale to accommodate a large number of nodes, making them suitable for large-scale deployments. They can also handle dynamic changes in network topology, such as the addition or removal of nodes.

Cost-effective: Since MANETs do not require any centralized infrastructure, they are often more cost-effective than traditional wired or wireless networks. They can also be used to extend the range of existing networks without the need for additional infrastructure.

Rapid Deployment: MANETs can be rapidly deployed in areas where infrastructure is not available, such as disaster zones or rural areas.

Disadvantages:

Security: MANETs are vulnerable to security threats, such as attacks by malicious nodes, eavesdropping, and data interception. Since the network is decentralized, there is no central authority to ensure the security of the network.

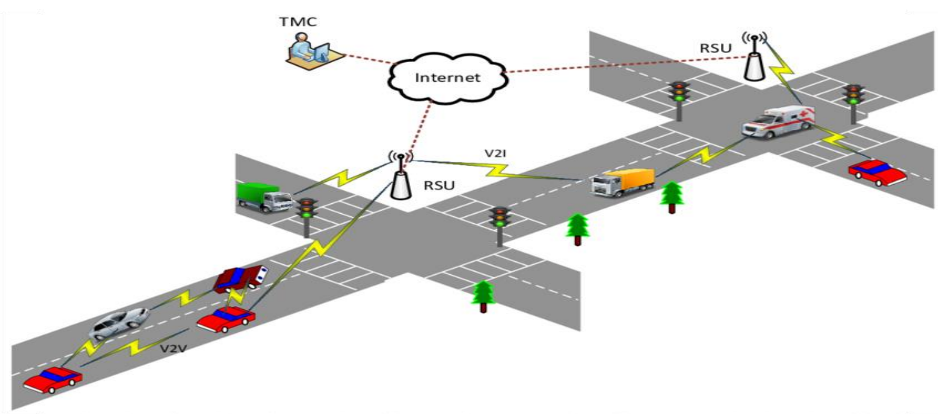
Reliability: MANETs are less reliable than traditional networks, as they are subject to interference, signal attenuation, and other environmental factors that can affect the quality of the connection.

Bandwidth: Since MANETs rely on wireless communication, bandwidth can be limited. This can lead to congestion and delays, particularly when multiple nodes are competing for the same channel.

Routing: Routing in MANETs can be complex, particularly when dealing with dynamic network topologies. This can result in inefficient routing and longer delays in data transmission.

Power Consumption: Since MANETs rely on battery-powered devices, power consumption can be a significant issue. Nodes may need to conserve power to extend the life of the battery, which can limit the amount of data that can be transmitted.

- **Vehicular Ad Hoc Networks (VANET) :**



VANET is similar to MANET in terms, that it also does not need any infrastructure for data transmission. VANET plays an important role in aspects of safe driving, intelligent navigation, emergency and entertainment applications .

It can be defined as an intelligent component of the transport system as vehicles are able to communicate with each other as well as roadside base stations, which are located at critical points of the road. Example :-Intersection and Construction Sites.

- **VANET** refers to a network created in an ad-hoc manner where different moving vehicles and other connecting devices come in contact over a wireless medium and exchange useful information to one another.
  A small network is created at the same moment with the vehicles and other devices behaving as nodes in the network.
- **VANET** comprises vehicle-to-vehicle & vehicle-to-infrastructure communications based on wireless local area network technologies.
- The distinctive set of candidate applications(e.g. Collision warning & local traffic information for drivers), resources(licensed spectrum,rechargeable power source), and environment(vehicular traffic flow pattern,privacy concerns)make the VANET a unique area of wireless communication.

❖ **Flying Adhoc network(FANET)**

- Flying Ad-hoc Networks (FANETs) are a specialized type of mobile ad-hoc network that are designed specifically for use in aerial vehicles, such as drones.

- They enable communication and coordination among a group of flying vehicles in a decentralized and self-organizing manner.

FANETs provide a flexible and reliable communication infrastructure for aerial vehicles, allowing for real-time data collection and transmission, as well as navigation and control.

They can operate in a standalone mode or can be connected to other networks, such as satellite or cellular networks, to provide extended communication capabilities.

Uses: FANETs have several potential uses in various fields such as:

- **Military and defense:** FANETs can be used for intelligence gathering, as well as for communication and coordination among military personnel and units.
- **Emergency response:** FANETs can be used to provide communication and coordination among emergency responders in the field, enabling effective response to natural disasters or other emergency situations.
- **Civil aviation:** FANETs can be used for air traffic management and control, as well as for communication and coordination among commercial and private aircraft.
- **Environmental monitoring:** FANETs can be used to collect and transmit data for environmental monitoring and research, such as for monitoring air and water quality, or for monitoring wildlife populations.
- **Agriculture:** FANETs can be used for precision agriculture, such as for monitoring crop health and for controlling crop-dusting drones.
- **Search and Rescue:** FANETs can be used to provide communication and coordination among search and rescue teams, enabling efficient and effective search and rescue operations.
- **Infrastructure inspection:** FANETs can be used for inspecting and monitoring large-scale infrastructure, such as bridges, buildings, and power lines.

- **Media and Entertainment:** FANETs can be used for live streaming and for capturing high-quality video and images for use in media and entertainment.
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❖ **Smart Phone Ad hoc Network (SPANC)**

- To create peer-to-peer networks without relying on cellular carrier networks,or traditional network infrastructure.

- ad-hoc network that utilizes smartphones as the primary nodes for communication.

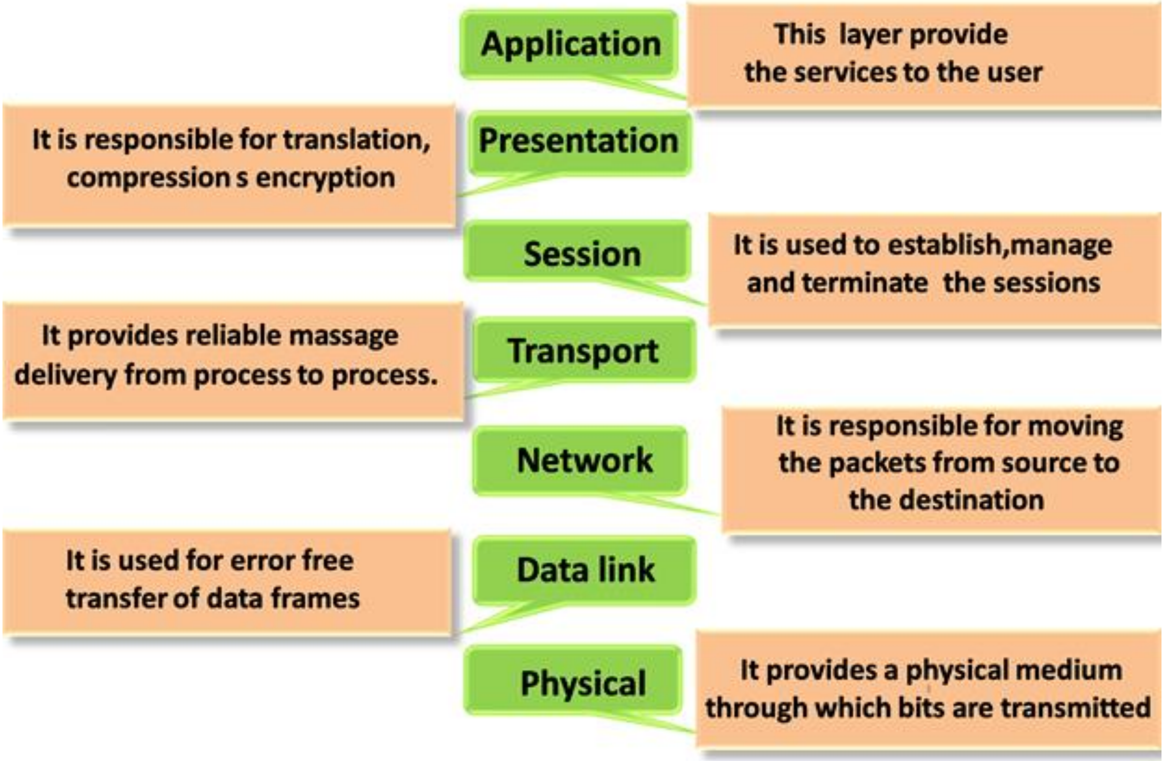
- In SPANC, smartphones can act as both routers and hosts, creating a decentralized network without the need for a central infrastructure.

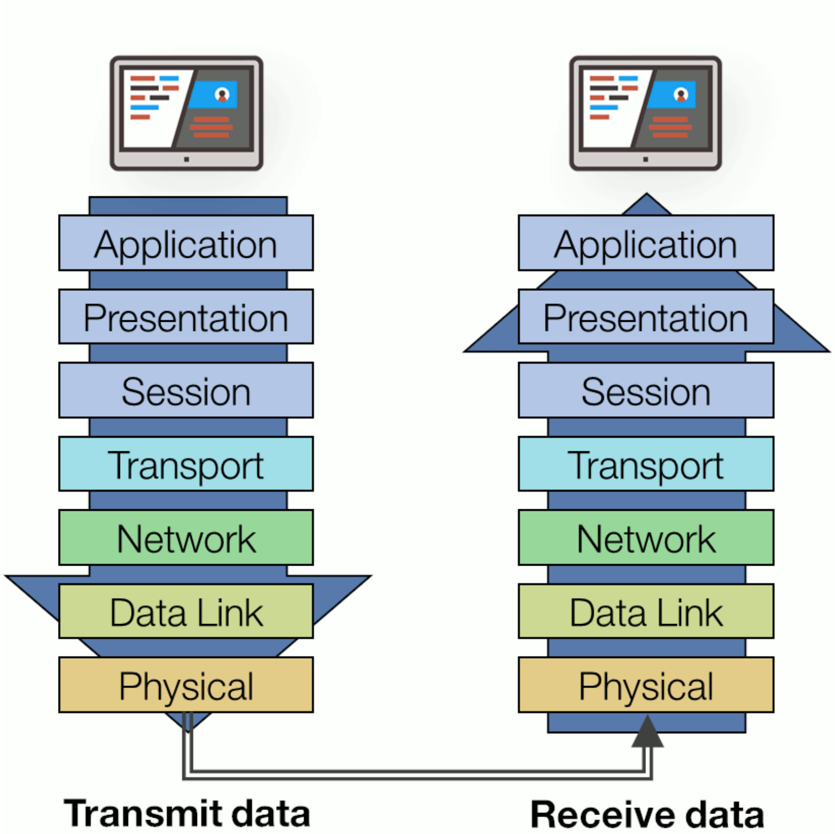
- This allows for increased flexibility and scalability in wireless communication, especially in emergency or disaster scenarios where traditional communication infrastructure may be unavailable.

- Some examples of SPANC applications include disaster response, search and rescue, and urban crowd management.

OSI MODEL

OSI stands for Open System Interconnection is a reference model that describes how information from a software application in one computer moves through a physical medium to the software application in another computer.


Application
This layer provide the services to the user
It is responsible for translation, compression s encryption
Presentation
Session
It is used to establish,manage and terminate the sessions
It provides reliable massage delivery from process to process.
Transport
Network
It is responsible for moving the packets from source to the destination
It is used for error free transfer of data frames
Data link
Physical
It provides a physical medium through which bits are transmitted



OSI consists of seven layers, and each layer performs a particular network function.
The OSI model was developed by the International Organization for Standardization (ISO) in 1984, and it is now considered as an architectural model for inter-computer communications.
The OSI model divides the whole task into seven smaller and manageable tasks. Each layer is assigned a particular task.

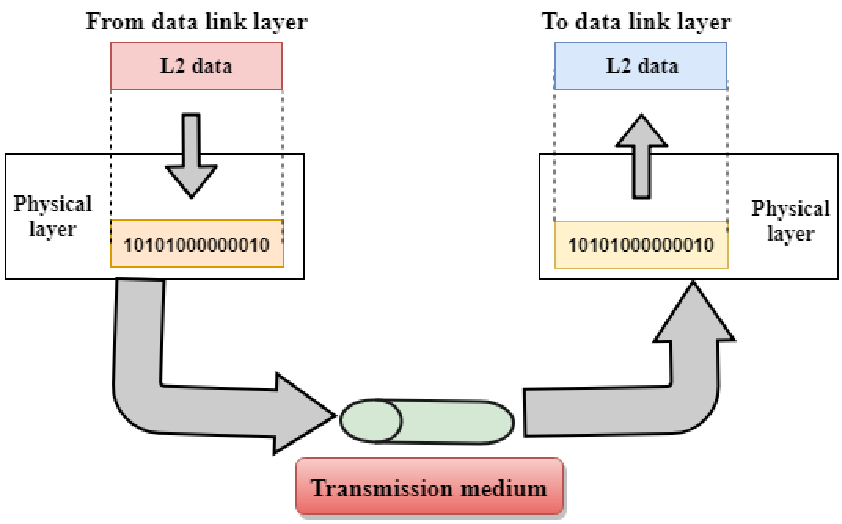
- There are the seven OSI layers. Each layer has different functions. A list of seven layers are given below:

    1. **Physical Layer**
    2. **Data-Link Layer**
    3. **Network Layer**
    4. **Transport Layer**
    5. **Session Layer**
    6. **Presentation Layer**

7. Application Layer

❖ Physical layer

- **The main functionality of the physical layer is to transmit the individual bits from one node to another node.**
- **It is the lowest layer of the OSI model.**
- **It establishes, maintains and deactivates the physical connection**



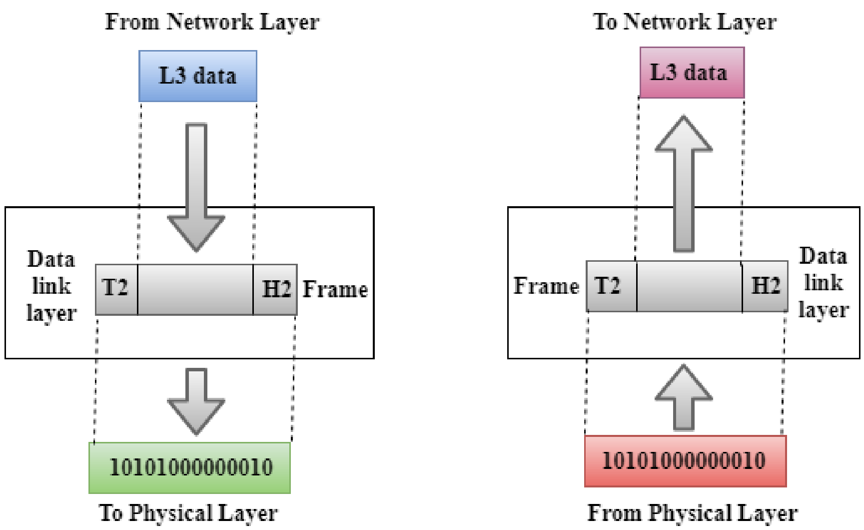
Functions of a Physical layer:

- **Line Configuration:** It defines the way how two or more devices can be connected physically.
- **Data Transmission**: It defines the transmission mode whether it is simplex, half-duplex or full-duplex mode between the two devices on the network.
- **Topology**: It defines the way how network devices are arranged.
- **Signals**: It determines the type of the signal used for transmitting the information.

❖ Data-Link Layer

- This layer is responsible for the error-free transfer of data frames.
- It defines the format of the data on the network.

- <u>**It contains two sub-layers**</u>:
- **Logical Link Control Layer**
    - It is responsible for transferring the packets to the Network layer of the receiver that is receiving.
    - It identifies the address of the network layer protocol from the header.
    - It also provides flow control.
- **Media Access Control Layer**
    - A Media access control layer is a link between the Logical Link Control layer and the network's physical layer.
    - It is used for transferring the packets over the network.



Functions of the Data-link layer:

<u>**Framing**</u>**: The data link layer translates the physical's raw bit stream into packets known as Frames. The Data link layer adds the header and trailer to the frame. The header which is added to the frame contains the hardware destination and source address.**

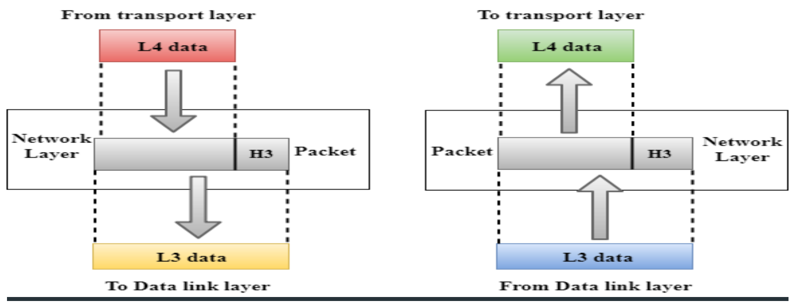
Physical Addressing: The Data link layer adds a header to the frame that contains a destination address. The frame is transmitted to the destination address mentioned in the header.

Flow Control: Flow control is the main functionality of the Data-link layer. It is the technique through which the constant data rate is maintained on both sides so that no data gets corrupted. It ensures that the transmitting station such as a server with higher processing speed does not exceed the receiving station, with lower processing speed.

Error Control: Error control is achieved by adding a calculated value CRC (Cyclic Redundancy Check) that is placed to the Data link layer's trailer which is added to the message frame before it is sent to the physical layer. If any error seems to occur, then the receiver sends the acknowledgement for the retransmission of the corrupted frames.

Access Control: When two or more devices are connected to the same communication channel, then the data link layer protocols are used to determine which device has control over the link at a given time.

- **Network Layer**



It is a layer 3 that manages device addressing, tracks the location of devices on the network.

It determines the best path to move data from source to the destination based on the network conditions, the priority of service, and other factors.

The Network layer is responsible for routing and forwarding the packets.

The protocols used to route the network traffic are known as Network layer protocols. Examples of protocols are IP and Ipv6.

Functions of Network Layer:

Internetworking: An internetworking is the main responsibility of the network layer. It provides a logical connection between different devices.

Addressing: A Network layer adds the source and destination address to the header of the frame. Addressing is used to identify the device on the internet.

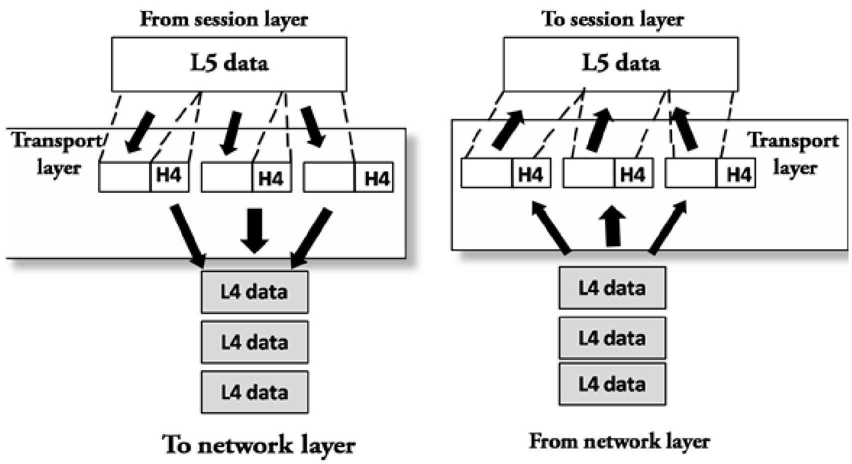
Routing: Routing is the major component of the network layer, and it determines the best optimal path out of the multiple paths from source to the destination.

Packetizing: A Network Layer receives the packets from the upper layer and converts them into packets. This process is known as Packetizing. It is achieved by internet protocol (IP).

❖ Transport Layer

The Transport layer is a Layer 4 ensures that messages are transmitted in the order in which they are sent and there is no duplication of data.

- The main responsibility of the transport layer is to transfer the data completely.
- It receives the data from the upper layer and converts them into smaller units known as segments.
- This layer can be termed as an **end-to-end** layer as it provides a point-to-point connection between source and destination to deliver the data reliably.



➔ <u>The two protocols used in this layer are:</u>

- **Transmission Control Protocol**
  - **It is a standard protocol that allows the systems to communicate over the internet.**
  - **It establishes and maintains a connection between hosts.**
  - **When data is sent over the TCP connection, then the TCP protocol divides the data into smaller units known as segments. Each segment travels over the internet using multiple routes, and they arrive in different orders at the destination. The transmission control protocol reorders the packets in the correct order at the receiving end.9**
- **User Datagram Protocol**
  - **User Datagram Protocol is a transport layer protocol.**
  - **It is an unreliable transport protocol as in this case the receiver does not send any acknowledgment when the packet is received, the sender does not wait for any**

acknowledgment. Therefore, this makes a protocol unreliable.

Functions of Transport Layer:

- **Service-point addressing:** Computers run several programs simultaneously due to this reason, the transmission of data from source to the destination not only from one computer to another computer but also from one process to another process. The transport layer adds the header that contains the address known as a service-point address or port address. The responsibility of the network layer is to transmit the data from one computer to another computer and the responsibility of the transport layer is to transmit the message to the correct process.
- **Segmentation and reassembly:** When the transport layer receives the message from the upper layer, it divides the message into multiple segments, and each segment is assigned with a sequence number that uniquely identifies each segment. When the message has arrived at the destination, then the transport layer reassembles the message based on their sequence numbers.
- **Connection control:** Transport layer provides two services Connection-oriented service and connectionless service. A connectionless service treats each segment as an individual packet, and they all travel in different routes to reach the destination. A

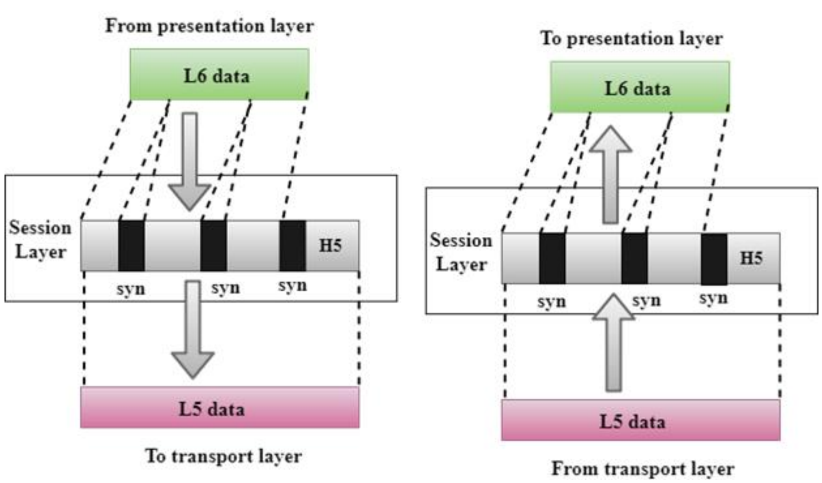
connection-oriented service makes a connection with the transport layer at the destination machine before delivering the packets. In connection-oriented service, all the packet travel in the single route.

- **Flow control:** The transport layer is also responsible for flow control but it is performed end-to-end rather than across a single link.
- **Error control:** The transport layer is also responsible for Error control. Error control is performed end-to-end rather than across the single link. The sender transport layer ensures that messages reach the destination without any error.

❖ Session Layer

is a layer 3 in the OSI model.

The Session layer is used to establish, maintain and synchronize the interaction between communicating devices.



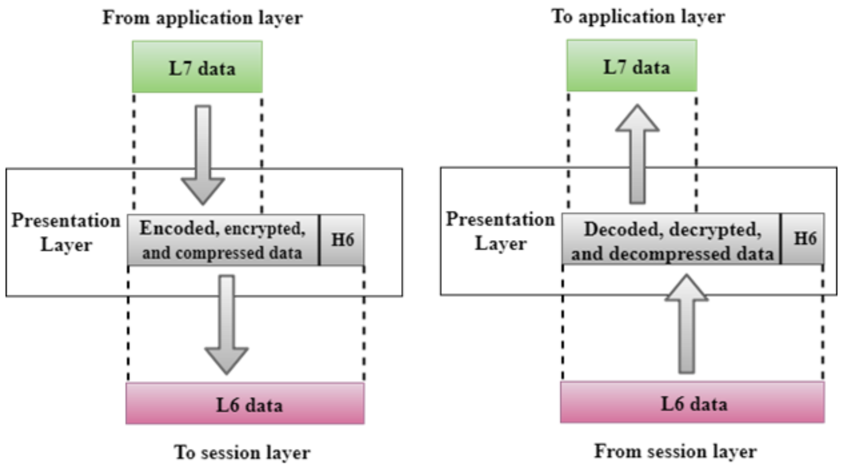
Functions of Session layer:

- **Dialog control:** Session layer acts as a dialog controller that creates a dialog between two processes or we can say that it allows the communication between two processes which can be either half-duplex or full-duplex.
- **Synchronization:** Session layer adds some checkpoints when transmitting the data in a sequence. If some error occurs in the middle of the transmission of data, then the transmission will take place again from the checkpoint. This process is known as Synchronization and recovery.
  - Example: If a system is sending a file of 800 pages, adding checkpoints after every 50 pages is recommended. This ensures that 50 page unit is successfully received and acknowledged.

❖ Presentation Layer

- A Presentation layer is mainly concerned with the syntax and semantics of the information exchanged between the two systems.
- It acts as a data translator for a network.
- This layer is a part of the operating system that converts the data from one presentation format to another format.

- The Presentation layer is also known as the syntax layer.

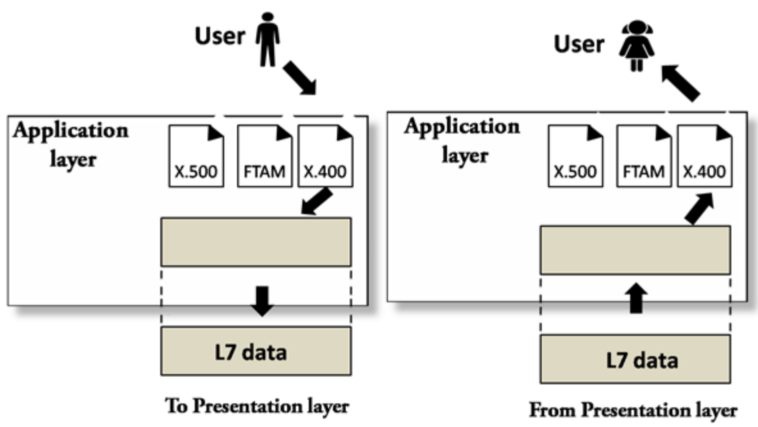


Functions of Presentation layer:

- **Translation:** The processes in two systems exchange the information in the form of character strings, numbers and so on. Different computers use different encoding methods, the presentation layer handles the interoperability between the different encoding methods. It converts the data from sender-dependent format into a common format and changes the common format into receiver-dependent format at the receiving end.
- **Encryption:** Encryption is needed to maintain privacy. Encryption is a process of converting the sender-transmitted information into another form and sends the resulting message over the network.
- **Compression:** Data compression is a process of compressing the data, i.e., it reduces the number of bits to be transmitted. Data compression is very important in multimedia such as text, audio, video.

❖ Application Layer

- An application layer serves as a window for users and application processes to access network service.
- It handles issues such as network transparency, resource allocation, etc.
- An application layer is not an application, but it performs the application layer functions.
- This layer provides the network services to the end-users.



Functions of Application layer:

- **File transfer, access, and management (FTAM):** An application layer allows a user to access the files in a, to retrieve the files from a computer and to manage the files in a remote computer.
- **Mail services:** An application layer provides the facility for email forwarding and storage.

- **Directory services**: An application provides the distributed database sources and is used to provide that global information about various objects.
    - Directory services are used to store, retrieve, and manage information about objects, such as user accounts, computer accounts, mail accounts, and information on resources available on the network.